Creativity Assessment Tool — Project Brief

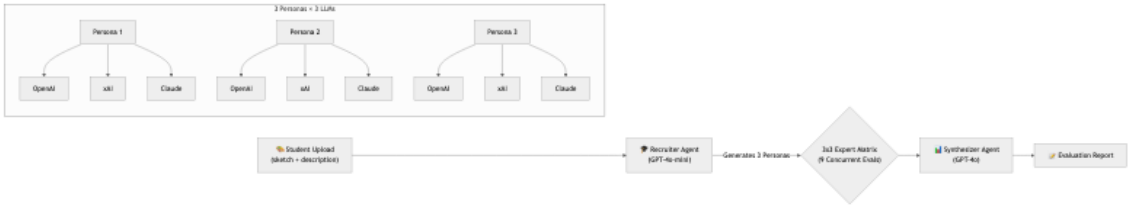
> *A multi-agent AI system that evaluates design creativity using the **Consensual Assessment Technique (CAT)** framework, built as a University of Oulu thesis project. It now features a powerful **3x3 Fan-Out matrix** architecture that yields 9 simultaneous evaluations.*

1. Problem Statement & Motivation

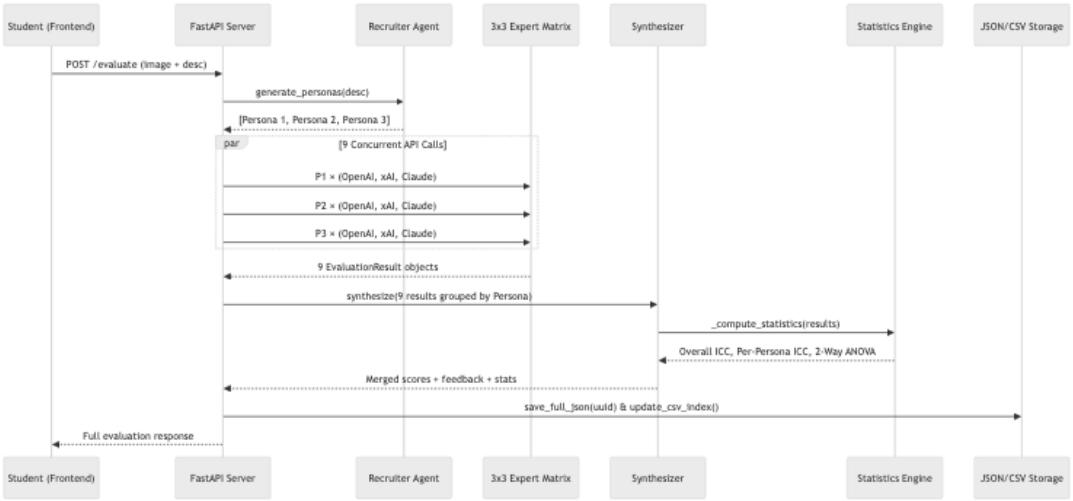
Evaluating creative design work is inherently subjective. The **Consensual Assessment Technique (CAT)**, the gold standard in creativity research, relies on multiple independent expert judges to score work, then measures **inter-rater reliability** to validate the assessment. This project replaces human judges with a matrix of **multiple LLM models role-playing multiple expert personas**, testing whether AI can replicate the CAT methodology at scale. The key research question is: *Can a 3x3 multi-agent LLM panel produce reliable, agreement-consistent creativity assessments?*

2. System Architecture

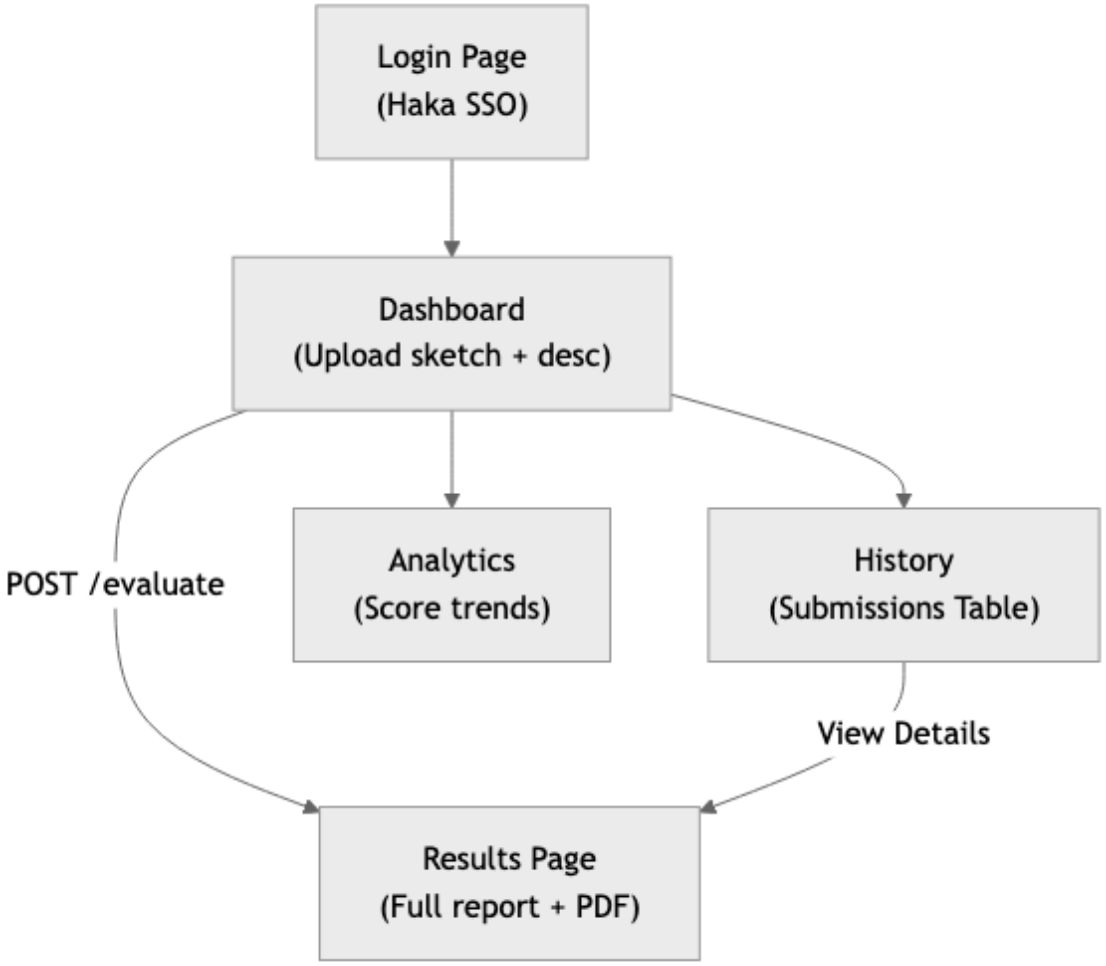
2.1 High-Level Pipeline: The 3x3 Fan-Out



2.2 Detailed Backend Data Flow



2.3 Frontend Page Flow



3. Pipeline Steps — Detailed Breakdown

Step 1: Recruiter Agent (`agents.py`)

Property	Detail
Model	`gpt-4o-mini`
Role	Assembles a tailored expert panel
Input	Assignment description
Output	3 `Persona` objects (name, title, sub_text, evaluation prompt)
Dynamic	Titles and focus areas change based on design domain

Step 2: Extract Panel: The 3x3 Matrix (`evaluators.py` & `creativity_judge.py`)

Once the 3 personas are generated, the orchestrator loops over them. For each persona, it fires off 3 asynchronous API requests to the LLM providers, yielding `3 Personas × 3 LLMs = 9 Evaluations` total.

Evaluator	Model	API	Fallback
OpenAI	`gpt-5.2`	OpenAI SDK	`gpt-4o`
xAI	`grok-4-1-fast-reasoning`	Custom base_url	None
Claude	`claude-3-5-sonnet-20241022`	Anthropic SDK	Strips markdown fences

All 9 evaluators enforce `temperature=0.1` and `response_format: json_object`.

Step 3: Synthesizer (`synthesizer.py`)

Component	Detail
Score Synthesis	Averages all 9 outcomes for the Composite Score
Instructor Feedback	`gpt-4o` writes Intro, Pivot, and Next Step
ICC Calculation	`pingouin` computes Overall ICC & Per-Persona ICC
Variance Analysis	Kendall's W for concordance and Dimension Variance to detect debate areas
Interpretation	LLM translates raw P-values and ICCs to plain-English badges

Step 4: Storage (`storage.py`)

- **Images** → `backend/data/images/{uuid}.{ext}`
- **Evaluations** → Full response saved as `backend/data/evaluations/{uuid}.json` to persist the 9 deep nested results.
- **Index** → Metadata (ID, overall score, date) appended to `backend/data/results.csv` for fast history populating.

4. Evaluation Rubric

All 9 parallel instances score the design on a **0–5 scale** across **6 dimensions**:

1. **Creativity** — General inventiveness
2. **Originality** — Uniqueness vs existing solutions
3. **Usefulness/Relevance** — Practical value
4. **Clarity** — Communication quality
5. **Level of Detail** — Depth and completeness
6. **Feasibility** — Technical/economic viability

5. Tech Stack

Backend

Layer	Technology	Purpose
Framework	**FastAPI**	Async REST API
Providers	`openai`, `anthropic`	LLM interactions
Statistics	`pandas`, `pingouin`	ICC(2) + Kendall's W + Variance Analysis

Storage	JSON + CSV + Filesystem	Persistence

Frontend

Layer	Technology	Purpose
UI	**React 19** + Vite	SPA Framework
Styling	**TailwindCSS 4**	Component styles & Dark Mode
Charts	**Recharts**	Radar/polar charts
PDF Export	**jsPDF**	Data-driven, layout-independent PDF rendering
HTTP	**Axios**	API calls

6. Frontend Pages — Detail

Results (`/results/:id`)

The centerpiece of the application. Highlights include:

- **Assessment Summary**: Image inspection modal (dark hover-reveal), radar consensus chart, dimension score breakdown, and Instructor Feedback.
- **Data-Driven PDF**: Downloads an A4 report directly using `jsPDF` without unstable DOM captures.
- **Expert Panel Accordion**: Groups the 9 evaluations by the 3 Personas. Expanding a persona reveals a 3-column grid comparing how OpenAI, xAI, and Claude interpreted that specific persona's prompt.
- **Statistical Reliability**: Displays Overall ICC, Per-Persona ICC, Kendall's W (Concordance), and Average Dimension Variance.

7. Statistical Methods — Deep Dive

The matrix architecture (3 Personas, 3 Models) enables robust multi-dimensional analysis:

- **Overall ICC(2)**: How well do all 9 evaluation paths agree? (Excellent ≥ 0.75, Poor < 0.40)
- **Per-Persona ICC**: Holding the persona constant, how well do the 3 AI models agree?
- **Kendall's W (Concordance)**: A normalized measure from 0.0 to 1.0 showing how consistently the 9 raters rank the 6 evaluated dimensions.
- **Dimension Variance**: Calculates the mathematical sample variance across the 9 scores for every dimension to identify exactly which aspect of the design caused the most debate among the AI panel.

8. Key Design Decisions

Decision	Rationale
3x3 Fan-Out	Provides statistically significant variance (N=9) while keeping latency low via concurrent execution.
Variance & Concordance	Tracking Kendall's W and Dimension Variance provides much clearer operational metrics on AI agreement than classical P-values.

Full JSON Storage	Relational DBs are overkill, but CSVs cannot easily store 9 deeply nested agent results. JSON files solve this perfectly.
jsPDF over window.print()	Browser print dialogs struggle with `oklch` modern CSS and dark mode overrides. jsPDF builds a clean, readable document directly from the result data.
Dark Mode First	The UI utilizes Tailwind dark mode extensively to provide a sleek, focused analytical environment.

9. Future Improvements

- **Human–AI Validation**: Compare the 3x3 AI matrix scores against true human expert judgments.
- **Customization Engine**: Let instructors define custom rubrics or force specific personas via the Dashboard UI.
- **Queue/Workers**: Move the heavy 9-request fan-out from synchronous HTTP requests to a Celery/Redis task queue with WebSockets for real-time progress bars.
- **Scale to N-x-N**: Allow dynamic configuration of `N personas x M models` based on varying computational budgets.